Full-Waveform Inversion for Ring-Array Ultrasound: Algorithm, Simulation Setup, and Abdominal CT Results

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October 27, 2026

Abstract

This note describes the time-domain full-waveform inversion (FWI) implemented in `invert_ring.py`, which recovers a two-dimensional sound-speed map from ring-array ultrasound traces. Applied to a CT-derived abdomen (1478–3476 m/s) with three-stage frequency continuation, it recovers the organ and muscle anatomy and the vertebral cortex, but leaves the vertebral body about 2000 m/s too slow because of cycle skipping in the bone. A logged coarse-stage rerun reduces the misfit 5.4× in 71 s on CPU and already shows the bone failing while the soft tissue converges.

1 Forward model

The forward model is the 2D constant-density acoustic wave equation, discretized with second-order finite differences in space and time (FDTD). Sound speed $c(\mathbf{x})$ is the only material property; density and attenuation are constant:

$$\frac{\partial^2 u}{\partial t^2} = c(\mathbf{x})^2 \, \nabla^2 u + s(t) \, \delta(\mathbf{x} - \mathbf{x}_{\text{tx}}). \tag{1}$$

- **Time stepping.** A leapfrog scheme advances the pressure, $u^{n+1} = 2u^n - u^{n-1} + \Delta t^2 c^2 L[u^n]$, where L is the 5-point Laplacian applied as a 3×3 convolution.
- **Stability.** The step obeys the 2D CFL bound $c_{\max} \, \Delta t \, \sqrt{1/\Delta x^2 + 1/\Delta y^2} < 1$. The code uses a CFL number of 0.45, with $c_{\max}$ the larger of the true medium maximum and the upper reconstruction bound.
- **Boundaries.** First-order Mur absorbing conditions on all four edges; corners average the x- and y-direction updates.
- **Sources and receivers.** A transmitter is a soft (additive) point source at its nearest grid node. Receivers sample pressure by bilinear interpolation, so element positions are not locked to pixel centres.
- **Execution.** Fields stay on one PyTorch device (CPU, CUDA or Apple MPS). Independent transmitter shots are advanced together as a batch, and on CUDA the step is compiled with `torch.compile`.

2 Simulation setup

A ring of point transducers surrounds the object; each element both transmits and receives. One element fires a chirp per shot and every element records, giving an $N \times N \times$ time dataset. Two configurations are used: the default phantom setup (200 mm field of view, 70 mm ring) and the abdominal setup, which scales the grid and ring to fit a 0.3 m-wide torso.

2.1 Acquisition and discretization

Table 1: Simulation parameters. Abdomen values (TotalSegmentator subject s1366) are listed per multiscale stage.

Parameter	Phantom setup (default)	Abdomen setup, per stage
Grid (cells)	401×401	$101^2 \rightarrow 201^2 \rightarrow 401^2$
Grid spacing Δx	0.50 mm	4.00 $\rightarrow$ 1.83 $\rightarrow$ 1.50 mm
Field of view	200 mm	404 $\rightarrow$ 367 $\rightarrow$ 602 mm
Ring radius	70 mm	167.7 $\rightarrow$ 168.3 $\rightarrow$ 168.5 mm
Elements	256 (pitch 1.7 mm)	128 (pitch 8.3 mm)
Coupling medium	water, 1500 m/s	water, 1480 m/s
Source pulse	Hann-windowed linear chirp, 100–250 kHz, 20 μs	25–75 kHz / 80 μs $\rightarrow$ 50–125 kHz / 40 μs $\rightarrow$ 100–250 kHz / 20 μs
Time step Δt (CFL 0.45)	$\approx$ 80 ns	326 $\rightarrow$ 149 $\rightarrow$ 122 ns
Record length	1.6$\times$ ring diameter / slowest speed	$\approx$ 371 μs (1135 $\rightarrow$ 2490 $\rightarrow$ 3039 steps)
Points per shortest wavelength	≈ 12	$\approx 4.8 \rightarrow 6.3 \rightarrow 3.9$
Inversion bounds on c	1400–2000 m/s	1400–3900 m/s

The abdomen time steps follow from the 3900 m/s upper bound, which sets the CFL limit. The record length is 1.6 times the time to cross the ring diameter at the slowest speed in the medium (fat, 1450 m/s). The grid is never finer than the CT pixels, so the final stage runs at the 1.5 mm CT spacing on a 602 mm field of view, most of it coupling water.

2.2 Abdominal sound-speed model from an open segmentation

The medium is one axial slice of a public, expert-segmented CT, converted by `totalseg_to_speed.py`. Organs, vessels, bowel, bone and the deep back muscles come from the segmentation; only pixels no mask covers are classified by Hounsfield units (HU).

1. **Data.** The *Small subset of TotalSegmentator Dataset* v2.0.1 [5, 4] holds 102 CT volumes resampled to 1.5 mm, each with masks of 117 structures (CC BY 4.0). `-fetch` downloads a single subject out of the 3.2 GB zip with HTTP range requests (44 MB for subject s1366).

2. **Subject choice.** Of the 54 subjects whose masks include the main upper-abdominal organs, the cleanest were screened by compressed bytes per voxel, then by the HU standard deviation inside the liver after removing a 5^3-voxel local mean. Subject s1366 has 14.7 HU of noise, against 140 HU for the noisiest (low-dose) scan, s0407, which was rejected.

3. **Slice choice.** Among slices showing all ten key structures (liver, spleen, both kidneys, pancreas, stomach, aorta, inferior vena cava, both deep back muscles), the one with the least gas inside the body is used: slice 259, at the T12/L1 level, with 2.5% gas.

4. **Tissue classes.** Masks are merged into the classes of Table 3. Unlabelled body pixels are classified by HU after a 5×5 median filter: fat (-190 to -30), gas (< -190) or other soft tissue. Gas, which the scalar fluid model cannot represent, becomes water; inside segmented solid organs low HU is treated as noise, not gas. The body is the largest connected region above -500 HU, which excludes the arms and the scanner table.

5. **Acoustic properties per class.** Each class points to one row of `abdominal_acoustic_parameters.csv` (Table 2), compiled from [1, 2, 3]. The script writes speed, density, power-law attenuation and B/A maps; the present solvers are lossless and constant-density, so only speed enters them. Gas is modelled as water rather than 343 m/s air: without a density contrast the scalar model cannot reproduce the near-total reflection at a gas boundary.

6. **Embedding.** `load_ct_ring_grid` centres the body, resamples it to each stage's grid with nearest-neighbour sampling, pads the outside with 1480 m/s water, and places the ring 10 mm outside the body with an 8-cell margin to the grid edge.

Table 2: Tissue acoustic properties (`abdominal_acoustic_parameters.csv`; values compiled from [1, 2, 3]). Attenuation follows $\alpha = \alpha_0 f^{\,y}$ with f in MHz.

Tissue	c (m/s)	ρ (kg/m^3)	α_0 (dB/MHz^y/cm)	y	B/A
Soft tissue (machine average)	1540	1000	0.5	1.1	7.0
Water / bile / urine	1480	1000	0.0022	2.0	5.0
Fat (visceral/subcutaneous)	1450	920	0.75	1.25	10.0
Liver	1570	1060	0.6	1.2	6.8
Kidney	1560	1050	0.9	1.0	7.2
Blood	1575	1050	0.15	1.3	6.1
Muscle (abdominal wall)	1580	1080	1.1	1.0	7.4
Spleen	1567	1054	0.55	1.2	6.8
Bowel gas / air	343	1.2	12.0	2.0	0.4
Bone / cortical rib	3750	1900	15.0	1.5	0.0

Table 3: Tissue classes of the s1366 slice (333×333 pixels at 1.5 mm) and the table row each uses.

Class	Source	Property row	c (m/s)	Share of body
Fat	HU	Fat	1450	31.9%
Other soft tissue	HU	Soft tissue average	1540	23.5%
Liver	mask	Liver	1570	21.8%
Digestive tract (stomach, small bowel, colon)	masks	Soft tissue average	1540	7.1%
Muscle (deep back muscles)	masks	Muscle	1580	4.3%
Gas, modelled as water	HU	Water	1480	2.5%
Bone (T11–L1 vertebrae, ribs)	masks	Bone	3750	2.3%
Kidney	masks	Kidney	1560	2.1%
Blood vessels (aorta, IVC, portal/splenic vein)	masks	Blood	1575	1.9%
Other organs (pancreas, adrenals)	masks	Soft tissue average	1540	1.8%
Spleen	mask	Spleen	1567	0.8%

The map spans 1450–3750 m/s. Soft-tissue contrasts are small (within 7% of water), while bone is 2.5 times faster than water: it dominates both the data and the difficulty of the inversion.

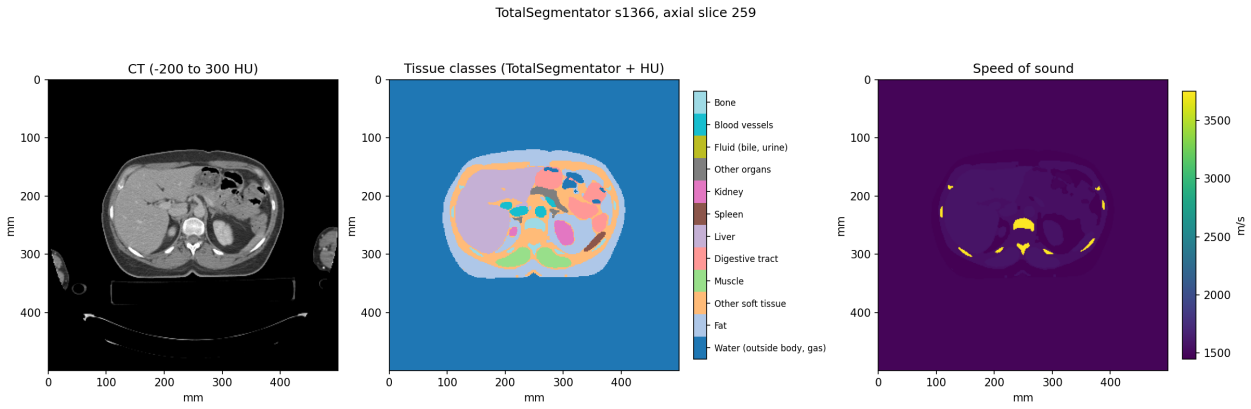


Figure 1: TotalSegmentator subject s1366, axial slice 259: CT (window −200 to 300 HU), merged tissue classes, and the resulting sound-speed map.

Earlier model. The first experiments (Section 6.1) used `ct_to_speed.py` on one slice of C4KC-KiTS case KiTS-00136 (512 × 512 at 0.684 mm). That pipeline has no anatomical labels: it thresholds HU into air, fat, soft tissue and bone, and splits superficial from deep tissue at 25 mm depth (fat into subcutaneous and visceral, soft tissue into muscle and organs), with representative speeds from Table 1 of [6] (1478–3476 m/s). The segmentation-based model replaces that depth rule with real organ, vessel and muscle boundaries.

3 Inversion algorithm

The inversion minimizes the least-squares misfit between simulated and observed traces over slowness squared $m = 1/c^2$, using an exact discrete adjoint for the gradient and first-order updates. Everything below is implemented in `invert_ring.py` and `fdtd2d/inversion/`.

3.1 Unknowns and constraints

- **Parameter.** $m = 1/c^2$ on every pixel inside an interior disk Ω of radius (ring radius − margin), with a 2-pixel margin by default. Outside Ω, c is fixed at the coupling-medium speed. The wave operator is linear in $c^2 = 1/m$, which keeps the gradient simple.
- **Bounds.** After every update m is projected onto $[1/c_{\max}^2,\ 1/c_{\min}^2]$ and the exterior is reset to the background.
- **Starting model.** Homogeneous coupling medium (1500 or 1480 m/s) inside the mask.

3.2 Objective

$$J(m) = \frac{1}{2} \sum_{s \in S} \sum_{r \neq s} \sum_{n} \left(d^{\text{pred}}_{s,r,n}(m) - d^{\text{obs}}_{s,r,n} \right)^2 + \frac{\alpha}{2} \left\| \nabla m \right\|^2_{\Omega}. \tag{2}$$

S is the set of active transmitters, r runs over receivers and n over time samples. The transmitter's own channel is excluded because a point source's direct pressure is a poorly conditioned datum. The Tikhonov term penalizes differences between neighbouring pixels inside Ω; its weight α (`--reg`) defaults to 0.

Observed data are synthetic: at the start of each stage the same solver simulates all N transmitters through the true medium, without noise.

3.3 Adjoint-state gradient

For each active shot the forward pass stores the full wavefield history u^n. The adjoint field λ is then stepped backward in time with the transposed operator $L(c^2\,\cdot)$, driven by the residuals scattered onto the grid by the transpose of the bilinear receiver sampling. The gradient with respect to c^2 accumulates as

$$g_{c^2} = \Delta t^2 \sum_n \lambda^n \odot L\big[u^{n-1}\big], \qquad \frac{\partial J}{\partial m} = -\frac{g_{c^2}}{m^2} + \alpha\,\nabla_m R(m). \tag{3}$$

The gradient is set to zero outside the mask. The adjoint reuses the forward Mur boundary update, which is not its exact transpose; the finite-difference gradient check (Section 5) bounds the effect of this.

3.4 Optimizers

Table 4: Optimizers available through `--optimizer`.

`--optimizer`	Update	Notes
`adam` (default)	Projected Adam on m, $\beta_1 = 0.9$, $\beta_2 = 0.999$	Step size `--adam-lr` in units of m ($\mathrm{s}^2/\mathrm{m}^2$): 10^{-9} default, 4×10^{-9} converges faster on coarse grids. Tolerates the noisy gradients of shot sub-sampling.
`cg`	Polak–Ribière nonlinear CG, Armijo backtracking, restart when $\beta < 0$	Deterministic; conjugacy is lost when shots are re-sampled each evaluation.
`gradient-descent`	Projected fixed step `--step-size`	Baseline.

`--model neural` replaces the per-pixel unknowns with the weights of a Fourier-feature coordinate network; the same adjoint gradient is passed through the network by the chain rule.

3.5 Shot sub-sampling

Each gradient evaluation uses K transmitters drawn at random without replacement (`--n-shots`, default 4; `--shot-seed` fixes the sequence), so one evaluation costs $2K$ wave solves instead of $2N$. Shots run in GPU batches of up to `--parallel-shots` (default 16).

3.6 Multiscale continuation

`--multiscale` runs three stages from low to high frequency on progressively finer grids (Table 1). Each stage regenerates its own observed data, and the final m of one stage is transferred to the next by bicubic interpolation. Starting at 25–75 kHz lengthens the period relative to the initial travel-time errors, which reduces cycle skipping: fitting a trace one period off, which traps single-scale inversions in a wrong model.

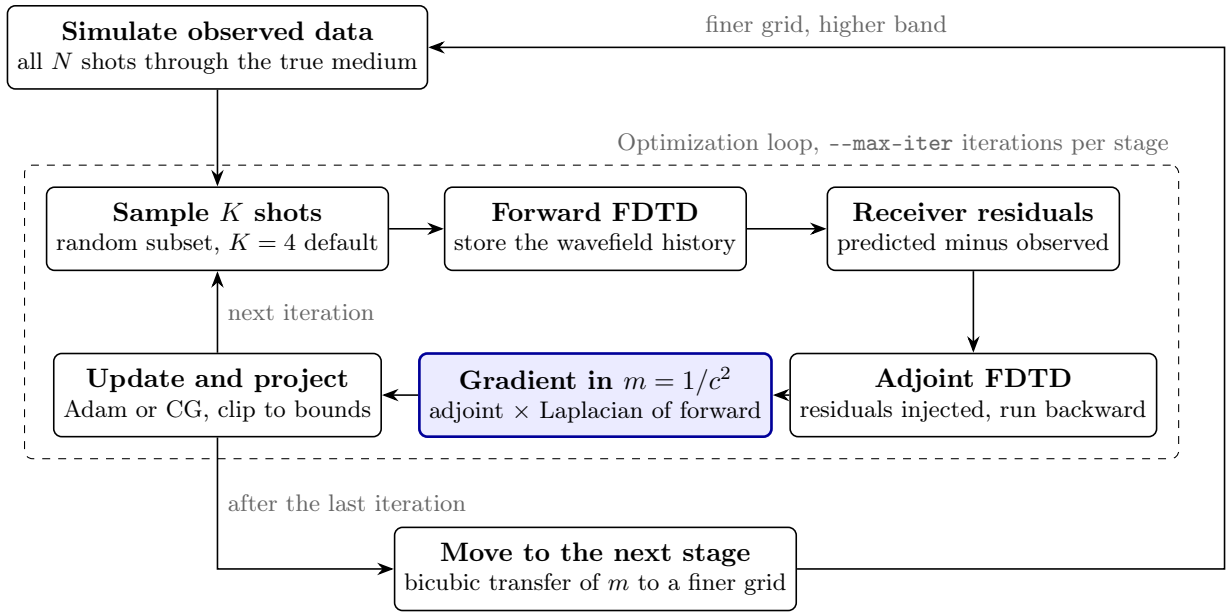


Figure 2: FWI loop inside each multiscale stage. Every iteration costs one forward and one adjoint solve per sampled shot; the three stages repeat the whole loop at 25–75, 50–125 and 100–250 kHz.

4 Algorithm at a glance

5 Implementation and verification

The adjoint gradient matches finite differences to second order, and the cost of one iteration is two wave solves per sampled shot plus the stored wavefield.

- **Batching.** `TorchFDTD2DBatch` advances up to `--parallel-shots` independent shots as one (batch, n_y, n_x) tensor; forward and adjoint passes share the same Laplacian convolution and Mur update. On CUDA both are compiled with `torch.compile(mode="reduce-overhead")`.
- **Memory.** The forward pass keeps every time step of every shot in the batch for the adjoint. In float32 this is $n_{\text{steps}} \times n_y \times n_x \times 4$ bytes per shot: about 1.3 GB for the 401^2 phantom grid and 1.95 GB for the 401^2 s1366 abdomen stage (3039 steps). The batch size is therefore memory-bound on the final stage; the s1366 run used `--parallel-shots 2` on a 17 GB machine.
- **Gradient check.** `python invert_ring.py --gradient-check --dtype float64` compares the adjoint directional derivative $g \cdot \delta m$ with central differences on a 41×41 grid with a 12 mm, 1800 m/s inclusion (Table 5).

Table 5: Adjoint gradient check (float64, run on September 28, 2026).

Step ε	Relative error
10^{-2}	3.5×10^{-3}
10^{-3}	3.4×10^{-5}
10^{-4}	3.4×10^{-7}

The error falls 100× per 10× smaller step, the expected behaviour for a correct gradient with a second-order difference; the approximate adjoint of the Mur boundary does not show at this level. A pytest suite (`tests/`) checks the neural parameterization's chain rule the same way on ring and arc problems.

6 Abdomen imaging results

6.1 Earlier experiment: KiTS-00136 with the HU-threshold model

The first abdomen runs used the `ct_to_speed.py` map of C4KC-KiTS case KiTS-00136 (categorical mapping, 1478–3476 m/s). Its stages ran at 4.63, 2.12 and 1.01 mm on 468, 425 and 406 mm fields of view, with a 194.4 mm ring of 64 elements (19 mm pitch). Three-stage FWI recovered the soft-tissue anatomy and the outer cortex of the vertebra, but left the vertebral body hollow, underestimated by up to about 2000 m/s.

6.1.1 Full multiscale reconstruction

- **Soft tissue.** Organ boundaries, the paraspinal muscles, the abdominal-wall muscle and the fat layers appear in the right places with roughly the right speeds; the difference map is near zero across most of the abdomen.
- **Bone.** The cortical rim of the vertebral body and the posterior elements are recovered, partly at the full 3476 m/s. The interior of the vertebral body stays near the background speed, the dark region in the difference map.
- **Artifacts.** Faint radial streaks fan inward from the ring. With 64 elements the pitch is about 19 mm, several wavelengths at 100–250 kHz, so the source and receiver sampling is sparse at the finest stage.
- **Convergence.** In the final stage J falls from about 5.9×10^3 to about 5.8×10^2 (roughly 10×, read from the plot), with a rise after the first step and the noise expected from random shot sub-sampling.

The command line for this run was not saved. Its setup is inferred from the figure (ring radius, 64 plotted elements, 50 Adam iterations per stage) and the code defaults.

6.1.2 Coarse-stage reproduction

To give logged numbers, the first stage was rerun on CPU on October 27, 2026 with

```
python invert_ring.py --ct-speed abdomen_acoustic.npz --coarse-only \
  --n-elements 64 --n-shots 4 --max-iter 50 --adam-lr 4e-9 \
  --c-min 1400 --c-max 3600
```

The full log is in `abdomen_coarse_stage_log.txt`.

The soft-tissue structures appear after a single coarse stage, but the vertebra goes the wrong way: speed falls to the lower bound where it should rise to 3476 m/s. A straight path through about 40 mm of bone arrives about 15 μs earlier than through water, more than half the 13 μs period at 75 kHz, so the coarse data are already cycle-skipped there. The later stages start from this hollow vertebra and only sharpen its rim, which matches Figure 3. The RMS error is dominated by the bone pixels; the difference map is close to zero over most of the soft tissue.

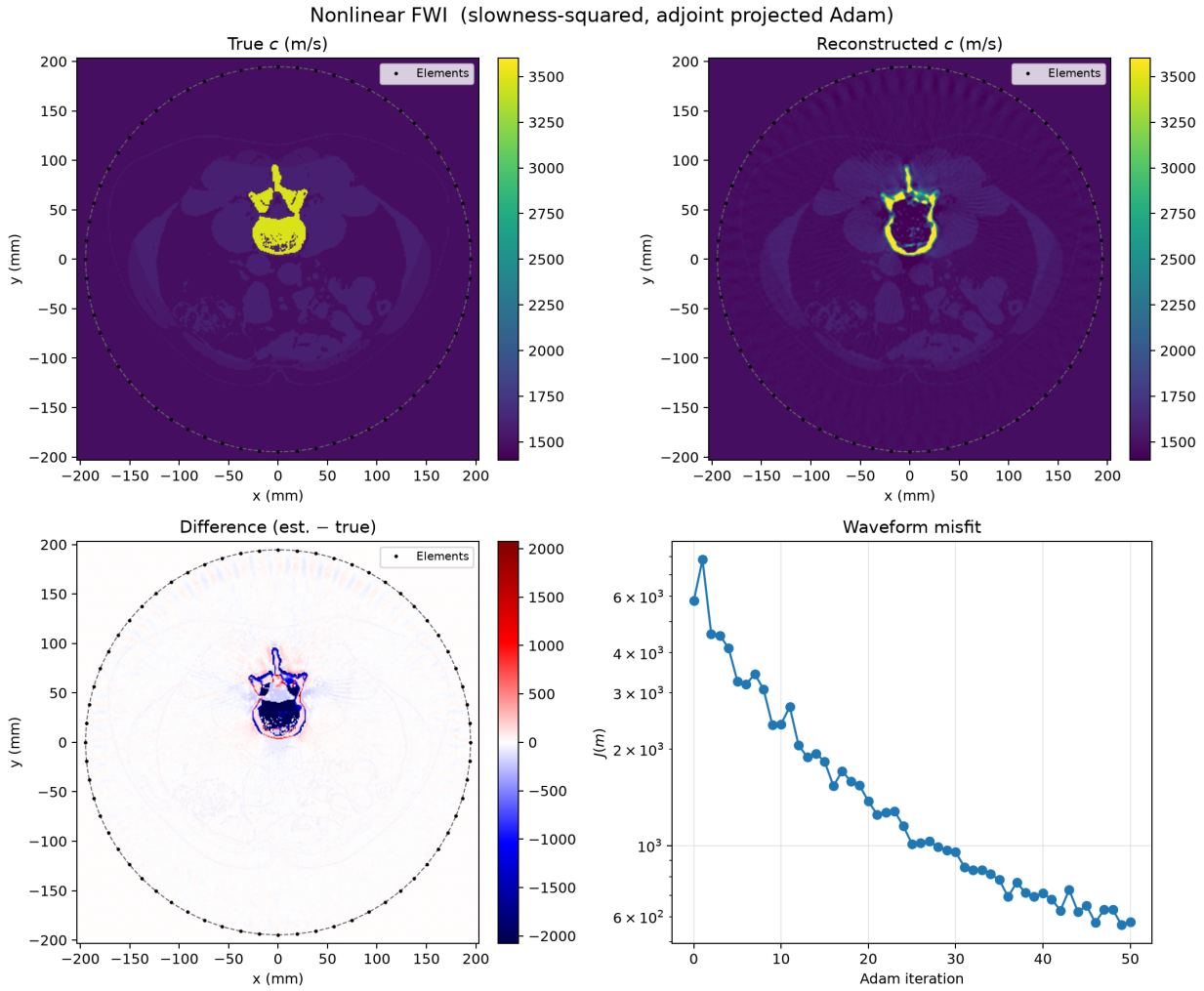


Figure 3: Multiscale FWI of the abdomen (`abdomenmultiscale.png`, committed September 26, 2026). True c, reconstruction after the 401^2 / 100–250 kHz stage, difference, and the final-stage waveform misfit over 50 Adam iterations.

7 Limitations and next steps

The main limits are the inverse crime, the bone, and the coarse sampling of the abdomen setup.

- **Inverse crime.** Observed and predicted data come from the same solver on the same grid, with no noise. Results are optimistic; data should be generated on a finer grid, or with an independent solver, plus noise.
- **Bone and cycle skipping.** The 2.35× speed contrast of the vertebra cycle-skips even at 25–75 kHz. Options: start below 25 kHz, add a travel-time or envelope misfit for the first stage, or begin from a bone-aware prior.
- **Grid resolution.** The abdomen stages run at about 4–6 points per shortest wavelength, low for a second-order scheme; numerical dispersion is masked only because data and model share the grid.

Table 6: Coarse-stage abdomen reproduction.

Quantity	Value
Grid / band	101×101 at 4.63 mm, 25–75 kHz
Time steps, Δt	1029, 409 ns
Misfit J, start $\to$ end	$7.35 \times 10^3 \to 1.35 \times 10^3$ (5.4$\times$ lower)
Interior RMS error in c	298.5 m/s
Interior c range, true	1478–3476 m/s
Interior c range, reconstructed	1400–1832 m/s
Wall time (8 CPU threads, float32)	71 s (observed data 2.1 s, optimization 68.6 s)

- **Element count.** 64 elements on a 194 mm ring (19 mm pitch) undersample the wavefield at 100–250 kHz and produce radial streaks. More elements, or more shots per iteration, should reduce them.
- **Physics.** Constant density, no attenuation and no shear waves. Bone and gas are only approximated by a scalar acoustic model.
- **Reproducibility.** The full multiscale run's settings were not logged; saving the command line and summary with each figure would close this gap.
- **Memory.** Storing the full forward wavefield (about 3 GB per shot on the 401^2 abdomen stage) caps the batch size; checkpointing would lift this.

Sources

- Repository scjoshi/SoundWaveSimulation: `invert_ring.py`, `fdtd2d/torch_backend.py`, `fdtd2d/inversion/` (CG, misfit, medium), `fdtd2d/ct_medium.py`, `ct_to_speed.py`, `fdtd2d/inversion/neural.py`.
- Abdomen figure `abdomenmultiscale.png`, commit `2de3791` (September 26, 2026); CT map figure `abdomen_acoustic.png`.
- CT slice: `ct-abdomen-c4kc-kits-instance.dcm`, C4KC-KiTS collection, The Cancer Imaging Archive.
- Coarse-stage reproduction: run on October 27, 2026; gradient check: run on September 28, 2026 (both CPU, PyTorch 2.6).

References

[1] F. A. Duck, *Physical Properties of Tissue: A Comprehensive Reference Book*. London: Academic Press, 1990.

[2] T. L. Szabo, *Diagnostic Ultrasound Imaging: Inside Out*, 2nd ed. Elsevier Academic Press, 2014.

[3] T. D. Mast, "Empirical relationships between acoustic parameters in human soft tissues," *Acoustics Research Letters Online*, vol. 1, no. 2, pp. 37–42, 2000.

[4] J. Wasserthal *et al.*, "TotalSegmentator: robust segmentation of 104 anatomic structures in CT images," *Radiology: Artificial Intelligence*, vol. 5, no. 5, e230024, 2023.

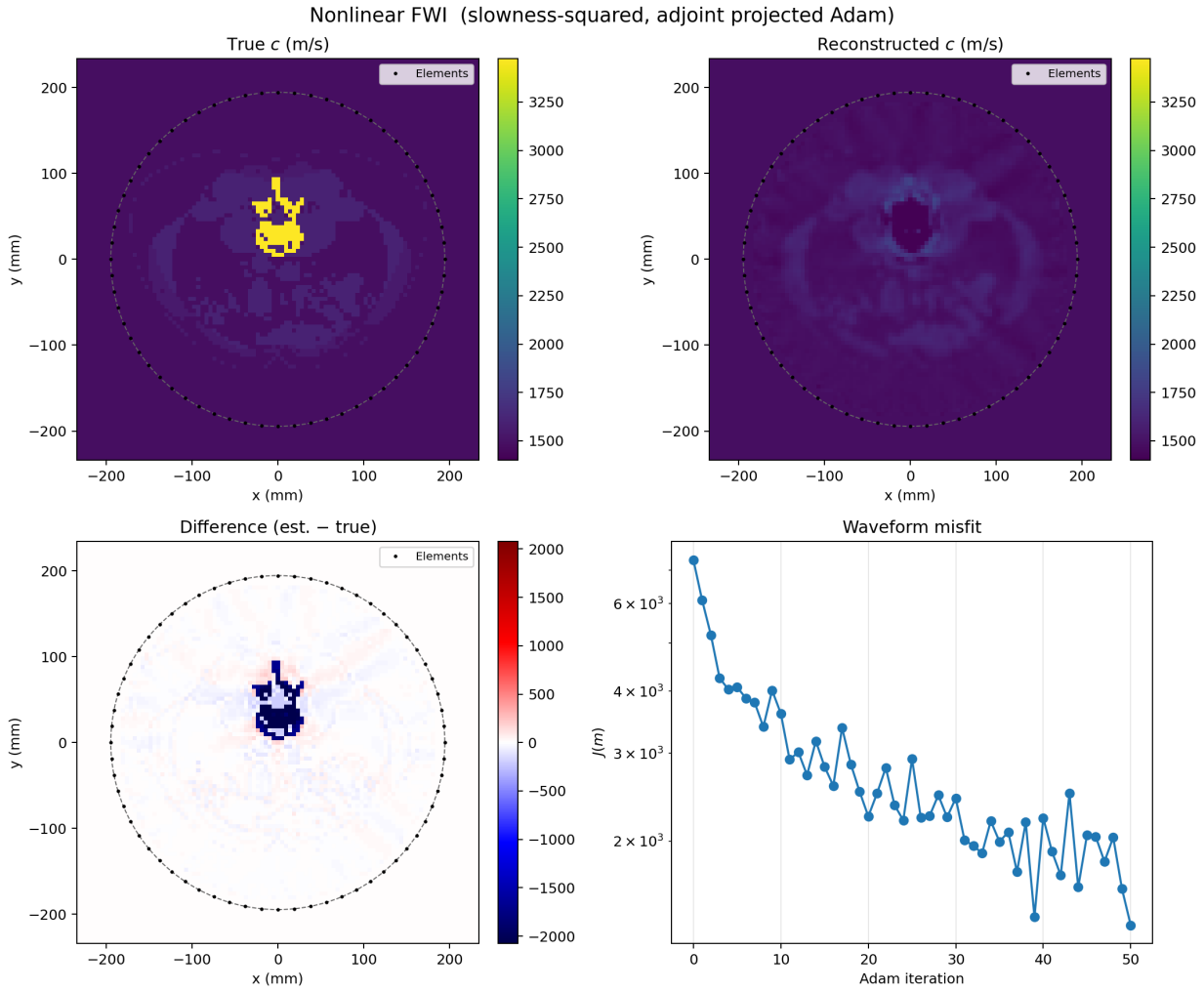


Figure 4: Coarse stage only (101^2, 25–75 kHz, 50 Adam iterations). Organ and muscle outlines are already visible; the vertebra region is driven down to the 1400 m/s lower bound.

[5] J. Wasserthal, "Small subset of TotalSegmentator Dataset," version 2.0.1, Zenodo, 2023, https://zenodo.org/records/10047263 (CC BY 4.0).

[6] M. O. Culjat, D. Goldenberg, P. Tewari, and R. S. Singh, "A review of tissue substitutes for ultrasound imaging," *Ultrasound in Medicine & Biology*, vol. 36, no. 6, pp. 861–873, 2010.